

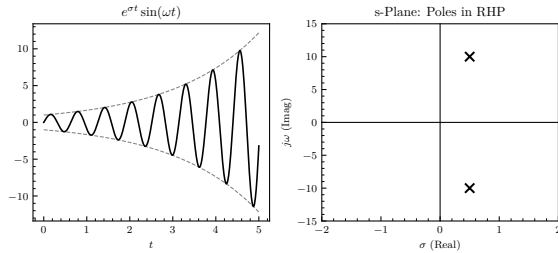
Control Cheatsheet

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Laplace Transform

Definition



The Laplace transform is a generalized Fourier transform used to simplify differential equations, either transforming PDEs into ODEs or converting ODEs into algebraic expressions. It is specifically designed to handle functions that grow exponentially at infinity. By definition, we have:

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty f(t)e^{-st} dt$$

In control theory, this transform is used to convert the time-domain equations of a dynamic system into a s-domain transfer function, making it easy to see if the natural response is unbounded (i.e., identifying poles with positive real parts).

Let $f(t) = e^{\sigma t} \sin(\omega t)$, $t \geq 0$, be a modulated signal, where σ is the growth rate and ω is the angular frequency. Taking the Laplace transform yields

$$F(s) = \frac{\omega}{(s - \sigma)^2 + \omega^2},$$

by setting the denominator to zero

$$s = \sigma \pm j\omega = 0.5 \pm j10,$$

because $\text{Re}(s) = \sigma > 0$ these poles lie in the Right-Half Plane (RHP), representing an unstable, exponentially growing oscillation of the time-domain signal.

Essential pairs

$f(t), t > 0$	$F(s)$
$\delta(t)$	1
u(t) (unit step)	$\frac{1}{s}$
t (unit ramp)	$\frac{1}{s^2}$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s+a}$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$
$e^{at} \sin(\omega t)$	$\frac{\omega}{(s+a)^2 + \omega^2}$
$e^{at} \cos(\omega t)$	$\frac{s+a}{(s+a)^2 + \omega^2}$

Properties

These equations represent the differentiation and integration properties of the Laplace transform,

$$\mathcal{L}\{\dot{f}\} = sF(s) - f(0),$$

$$\mathcal{L}\{\ddot{f}\} = s^2 F(s) - sf(0) - \dot{f}(0),$$

$$\mathcal{L}\left\{\int_0^t f d\tau\right\} = \frac{F(s)}{s},$$

In control theory we almost always assume zero initial conditions $f(0) = 0$.

If and only if all poles of $sF(s)$ are in the Left-Half Plane (LHP) we can verify by the final value theorem a steady-state output,

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$$